# Feng Tian

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## RESEARCH INTERESTS

- Computational Imaging
- Computer Vision
- Machine Learning
- Neuroengineering
- Optical Design
- Microfabrication

My research focuses on machine learning for computational imaging and optical design, with emphasis on deep learning, optimization-based reconstruction algorithms, and inverse design of diffractive optical elements. I develop lensless imaging devices for implantable photography and microscopy applications. My overall research goal is to address the challenge of AI-powered 3D sensing with compact device for machine vision and neural science applications.

## EDUCATION AND EMPLOYMENT

**University of California, Davis** Davis, CA
**Postdoctoral Scholar** February 2025 - Present

**University of California, Davis** Davis, CA
**Ph.D. in Electrical and Computer Engineering** September 2018—December 2024

**Ams OSRAM** Boise, ID
**Research Intern** June 2022—July 2022

- Conducted design and FDTD simulations on metastructure optical elements.
- Developed computational imaging algorithms for object reconstruction through deep learning.

**Nanophotonics and Nanofabrication Group, University of Michigan** Ann Arbor, MI
**Research Intern** May 2018—August 2018

- Conducted proof-of-concept experiment on characterizing optoelectronic tactile shear stress sensor.
- Developed an autofocusing and mechanical control algorithm for optical microscope.

**University of Michigan** Ann Arbor, MI
**M.S. in Electrical and Computer Engineering** September 2016—April 2018

**University of Electronic Science and Technology of China** Chengdu, Sichuan
**B.S. in Electrical and Computer Engineering** September 2012—June 2016

- Thesis: All-Optical Modulation Format Conversion Based on Nonlinear Dynamics of Optical Injected Semiconductor Laser

## AWARDS

Richard and Joy Dorf Graduate Student Award May 2023
UC Davis Dissertation fellowship April 2023
UC Davis Research Award April 2022
UC Davis Graduate Student Travel Award December 2021

## JOURNAL PUBLICATIONS

- **Feng Tian**, Weijian Yang, "Photonics Breakthroughs 2025: Mask-based Integrated Miniscope for Large-scale Volumetric Imaging", ***IEEE Photonics Journal*** 18, 3900408 (2026). [Link]
- **Feng Tian**, Weijian Yang, "Unrolled Richardson-Lucy deconvolution network with partially connected layers in computational microscopy", ***Optics Express*** 34, 18676-18689 (2026). [Link]
- **Feng Tian**, Ben Mattison, Weijian Yang, "DeepInMiniscope: Deep learning–powered physics-informed integrated miniscope", ***Science Advances*** 11, eadr6687 (2025). [Link]
- Yunyang Li, Zixiao Zhang, **Feng Tian**, Yryx Y. Luna-Palacios, Israel Rocha-Mendoza, Weijian Yang, "V-shaped PSF for 3D imaging over an extended depth of field in wide-field microscopy", ***Optics Letters*** 50, 383-386 (2025). [Link]
- **Feng Tian**, Weijian Yang, "Learned lensless 3D camera", ***Optics Express*** 30, 34479-34496 (2022). [Link]
- **Feng Tian**, Junjie Hu, Weijian Yang, "GEOMScope: Large field-of-view 3D lensless microscopy with low computational complexity", ***Laser & Photonics Reviews*** 15, 2100072 (2021). [Link]
- Jingyang Sui, Kunook Chung, **Feng Tian**, Pei-Cheng Ku, "A tensorial shear stress sensor based on light-emitting GaN nanopillars", ***Applied Physics Letters*** 115, 021103 (2019). [Link]

## CONFERENCES PROCEEDINGS

- Feng Tian, Weijian Yang, Ben Mattison "Physics-aware lensless miniscope for neuronal imaging (Invited Paper)", Proc. SPIE PC13333, Computational Optical Imaging and Artificial Intelligence in Biomedical Sciences II (2026).
- Feng Tian, Weijian Yang, "Compact 3D fluorescence microscopes using a single layer of microlens array (Invited Paper)", Proc. SPIE PC13330, High-Throughput Biophotonics: Imaging, Spectroscopy, and Beyond X (2025).
- Yunyang Li, Shu Guo, Ben Mattison, Shing-Jiuan Liu, Feng Tian, Weijian Yang, "High-speed short-line-excitation two-photon microscopy to image brain activity (Invited Paper)", Proc. SPIE PC13303, Neural Imaging and Sensing (2025).
- Feng Tian, Weijian Yang, "Physics aware lensless miniscope for neuronal imaging", Proc. SPIE PC13333, Computational Optical Imaging and Artificial Intelligence in Biomedical Sciences II (2025).
- Ben Mattison, Shing-Jiuan Liu, Feng Tian, Weijian Yang, "High-speed miniaturized two-photon microscopy with elliptical beam excitation", Proc. SPIE PC12828, Neural Imaging and Sensing 2024 (2024).
- Feng Tian, Weijian Yang "Miniaturized and Thin Microscope with Learnable Real-Time Reconstruction", Conference on Lasers and Electro-Optics, Optica Publishing Group, ATu3Q.3 (2023).
- Yunyang Li, Zixiao Zhang, Feng Tian, Yryx Luna-Palacios, Israel Rocha-Mendoza, Weijian Yang, "V-Shape PSF for 3D Wide Field Microscopy", Conference on Lasers and Electro-Optics, Optica Publishing Group, ATu4Q.3 (2023).
- Ben Mattison, Shing-Jiuan Liu, Feng Tian, Weijian Yang, "High Speed Miniaturized Multiphoton Microscopy With Elliptical Beam Excitation", Biophotonics Congress: Optics in the Life Science, Optica Publishing Group, JTu4B.33 (2023).
- Feng Tian, Weijian Yang, "Miniaturized and Thin 3D Microscope With Learnable Real-Time Reconstruction", Biophotonics Congress: Optics in the Life Science, Optica Publishing Group, BM3B.3 (Student Paper Finalist) (2023).
- Feng Tian, Weijian Yang, "Miniaturized fluorescence 3D microscope with deep learning reconstruction", Proc. SPIE 12385, Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXX (2023).
- Feng Tian, Weijian Yang, "Computational 3D imaging with micro optics and data-driven deep learning reconstruction", Proc. SPIE 12019, AI and Optical Data Sciences III (2022).
- Feng Tian, Weijian Yang, "3D Camera using a Single Layer of Microlens Array and a Learned Reconstruction Algorithm", Conference on Lasers and Electro-Optics, Optica Publishing Group, JW3A.7 (2022).

- Feng Tian and Weijian Yang, "Miniaturized and Thin 3D Microscope with Fast Reconstruction," Conference on Lasers and Electro-Optics, Optica Publishing Group, SM5O.3 (2022).
- Feng Tian, Junjie Hu, and Weijian Yang, “Large field-of-view 3D Imaging using Random Microlenses”, Conference on Lasers and Electro-Optics, Optica Publishing Group, SW2D.3 (2021).
- Feng Tian, Junjie Hu, and Weijian Yang “Large field-of-view three-dimensional imaging with low computational complexity using random microlens array”, Proc. SPIE 11649, Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXVIII (2021)
- Feng Tian, Junjie Hu, and Weijian Yang, “Three-dimensional Imaging with a Single Layer of Random Microlens Array”, Conference on Lasers and Electro-Optics, Optica Publishing Group, STh3M.3 (2020).
- Kunook Chung, Feng Tian, Jingyang Sui and Pei-Cheng Ku, “Tensorial Shear Stress Sensing Using Elliptically-Shaped Nanopillar Light-Emitting Diodes”, Conference on Lasers and Electro-Optics, Optica Publishing Group, AM4I.4 (2019).

## TEACHING EXPERIENCE

**University of California, Davis** Davis, CA

**Teaching Assistant**

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|---|---|
| EEC 100 – Circuits II | Winter 2024 |
| EEC 100 – Circuits II | Fall 2023 |
| EEC 130A – Electromagnetics I | Fall 2022 |
| EEC 243 – Optical Imaging and Microscopy | Winter 2021 |
| EEC 140A – Principles of Device Physics I | Fall 2019 |
| EEC 140B – Principles of Device Physics II | Spring 2019 |

## PROFESSIONAL SERVICES

**Journal Reviewer**

- Laser & Photonics Reviews
- Optics Express
- Optics Letters
- IEEE Transactions on Signal Processing
- Applied Physics Letters
- Applied Optics